

CRISPR Informatics Course

Design and *in silico* analysis of CRISPR and CRISPR Base Editors gRNAs

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We will go through:

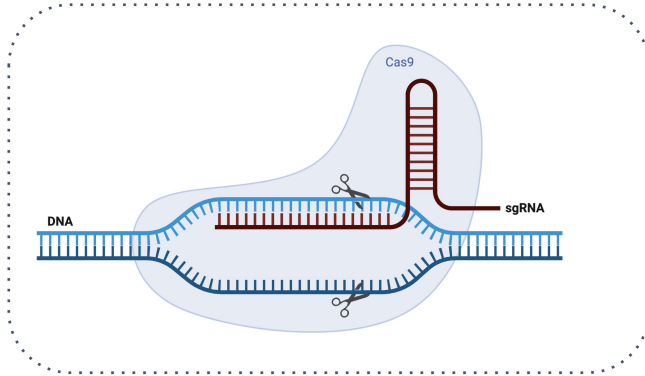
i. guide RNAs for CRISPR systems

ii. gRNA libraries

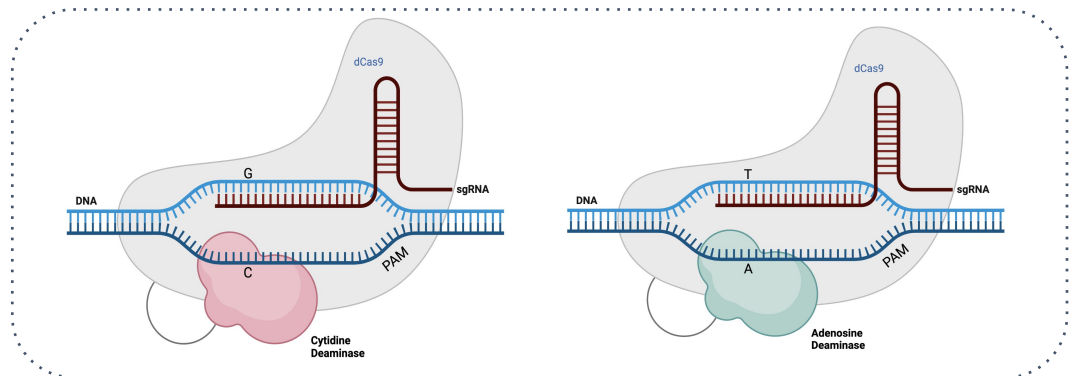
iv. Practical session to design Base Editor gRNAs

CRISPR Technologies

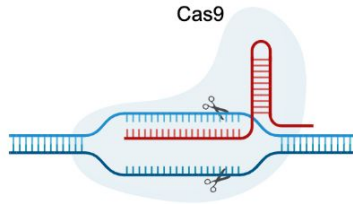
CRISPR – Cas9 System



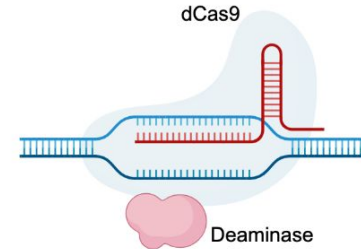
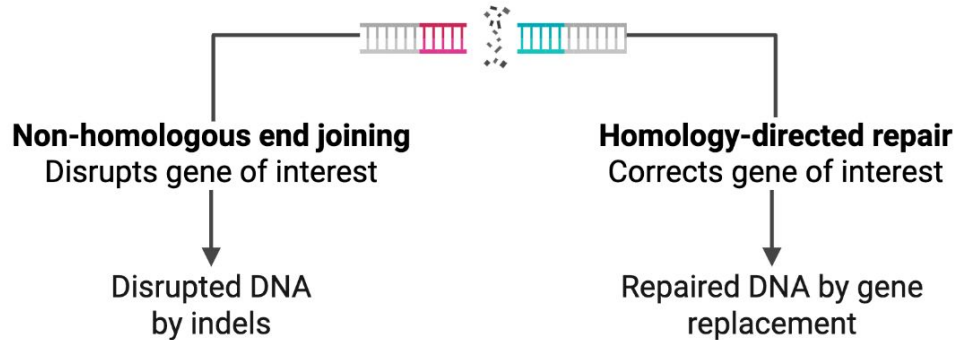
CRISPR Base Editing System



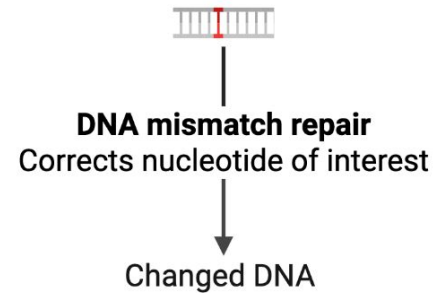
CRISPR Technologies



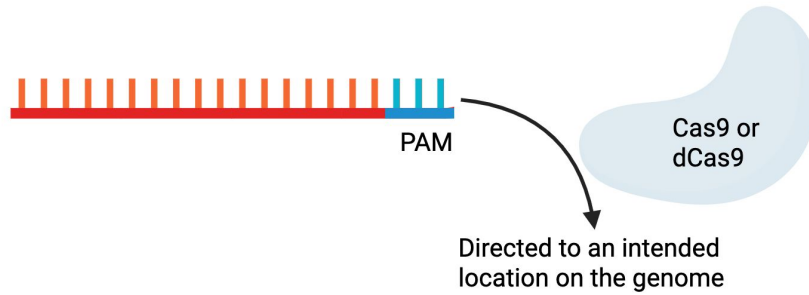
Double-stranded break



Nucleotide Deamination



gRNAs



Things to consider:

- > On target and off targets
- > GC content
- > Nucleotide repetition
- > Motifs
- > Efficacy (affected by cell type or surrounded sequences etc)
- > Based on reference genome

gRNA Library

For CRISPR-KO experiments, there are genome-wide screens across hundreds of models



cancer
dependency
map



depmap

Article | [Open access](#) | Published: 21 December 2018

Optimized libraries for CRISPR-Cas9 genetic screens with multiple modalities

[Kendall R. Sanson](#), [Ruth E. Hanna](#), [Mudra Hegde](#), [Katheri Sullender](#), [Emma W. Vaimberg](#), [Amy Goodale](#), [David E. R.](#)

[Nature Communications](#) **9**, Article number: 5416 (2018)

134k Accesses | **843** Citations | **49** Altmetric | [Metrics](#)

Short Report | [Open access](#) | Published: 21 January 2021

Minimal genome-wide human CRISPR-Cas9 library

[Emanuel Gonçalves](#), [Mark Thomas](#), [Fiona M. Behan](#), [Gabriele Picco](#), [Clare Pacini](#), [Felicity Allen](#), [Alessandro Vinceti](#), [Mamta Sharma](#), [David A. Jackson](#), [Stacey Price](#), [Charlotte M. Beaver](#), [Oliver Dovey](#), [David Parry-Smith](#), [Francesco Iorio](#), [Leopold Parts](#), [Kosuke Yusa](#) & [Mathew J. Garnett](#)

[Genome Biology](#) **22**, Article number: 40 (2021) | [Cite this article](#)

16k Accesses | **62** Citations | **16** Altmetric | [Metrics](#)

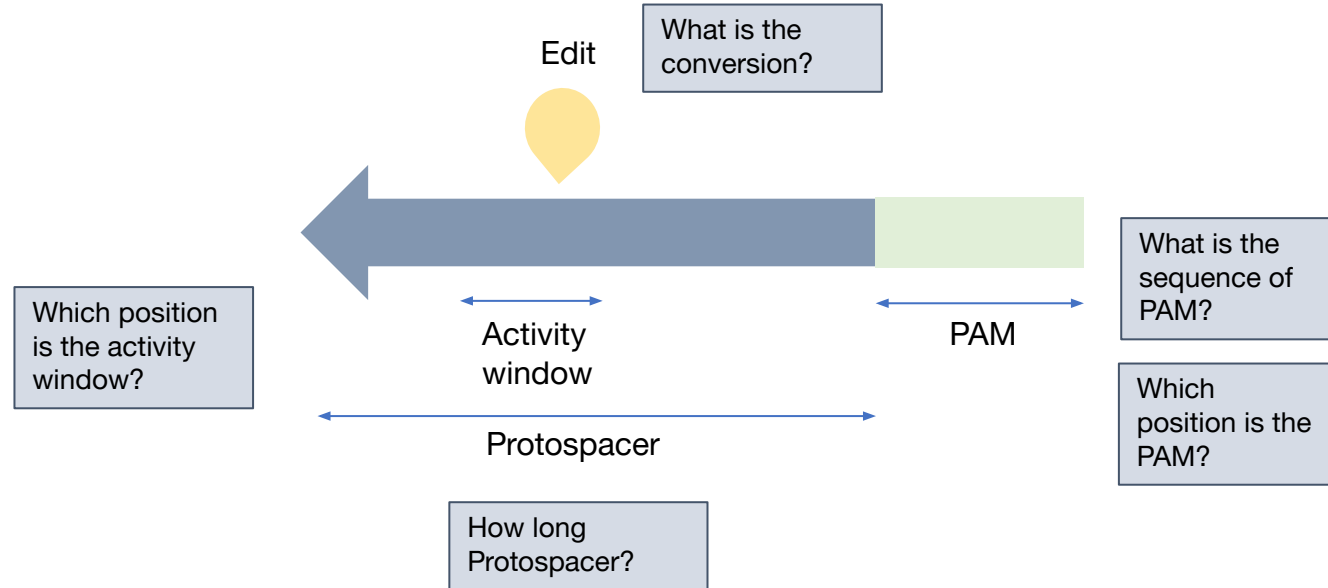
gRNA Library

Wellcome Sanger Institute Genome Editing (WGE)

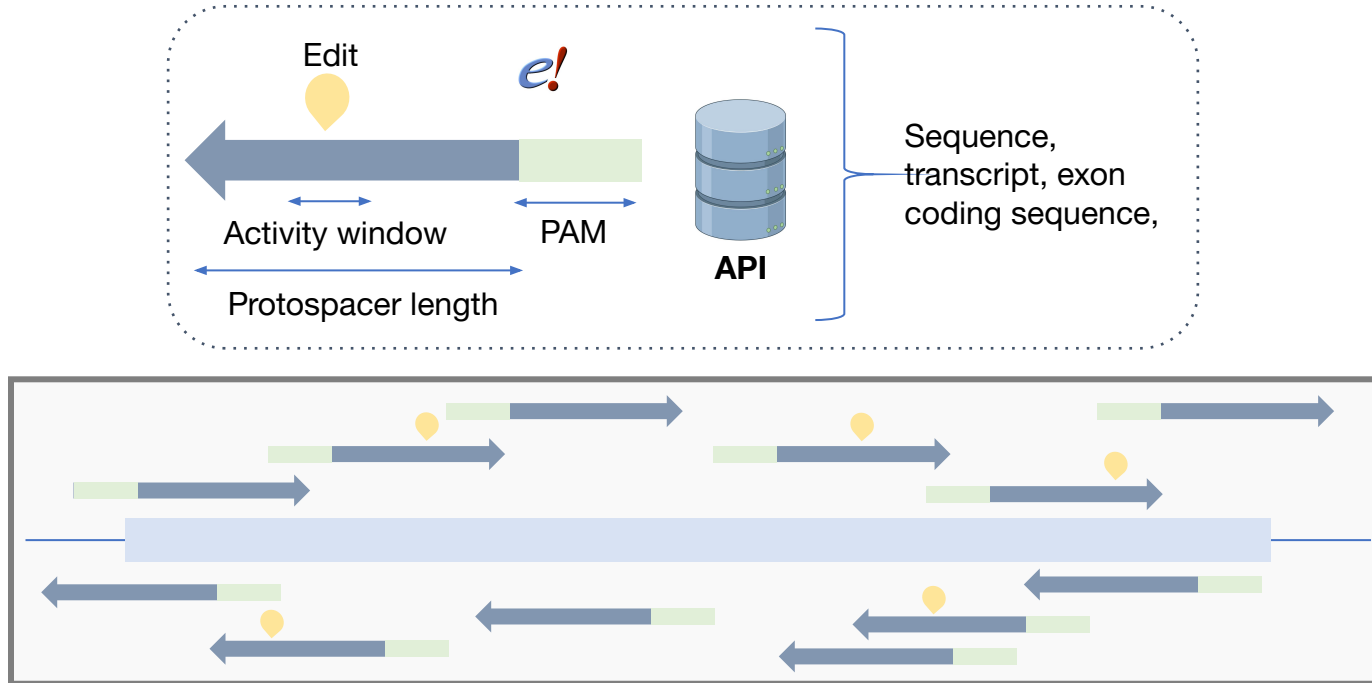


https://wge.stemcell.sanger.ac.uk/find_crisprs

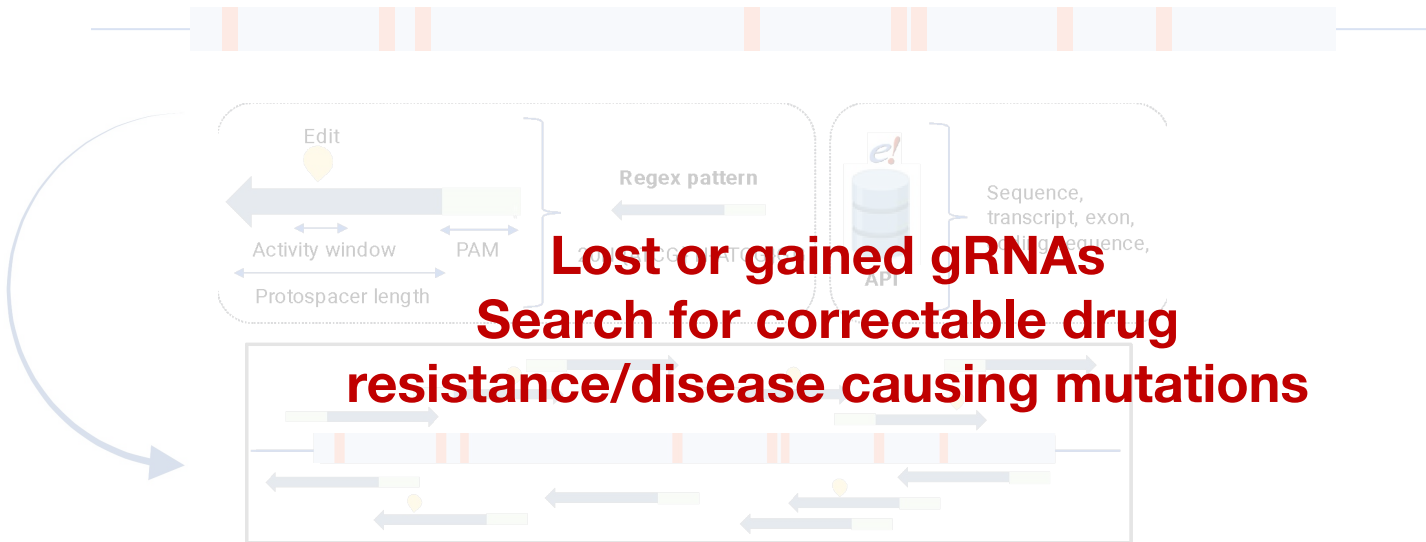
gRNA Library design for Base Editors with BEstimate



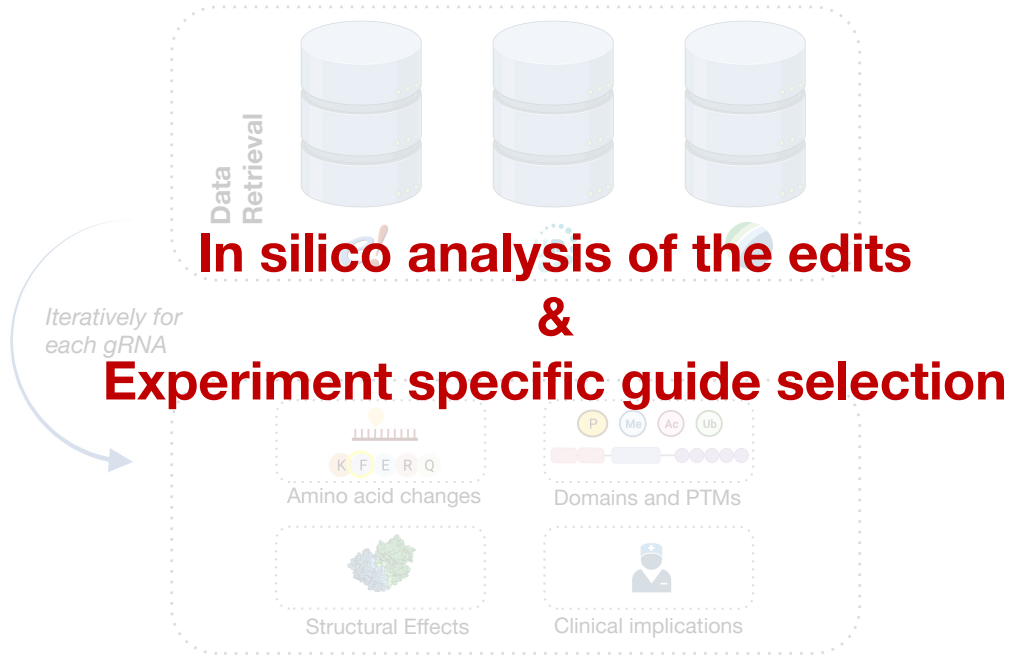
Step – 1 : Finding the gRNAs



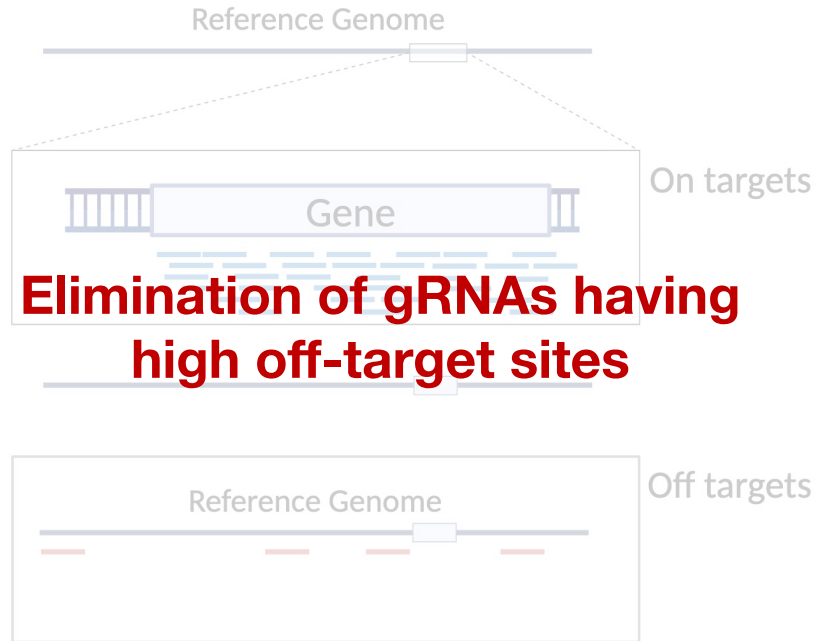
Step – 1 : Finding the gRNAs on mutated genes



Step – 2 : gRNA Annotation



Step – 3 : Off-target Annotation



Other tools to discover

- > [PnB Designer](#): gRNA design for DNA sequences and variants
- > [beditor](#): gRNA design for variants
- > [BE-Designer](#): gRNA design for sequences
- > [SNP-CRISPR](#): gRNA design for variants
- > [BE-FF](#): gRNA design for variants
- > [crisprVerse](#): gRNA design for DNA sequences

Predicting base editing outcomes using position-specific sequence determinants

FORECasT-BE for gRNA efficiency

BE-Hive for gRNA efficiency



BE-Hive

[SINGLE MODE](#) · [BATCH MODE](#) · [EDITOR EFFICIENCIES](#) · [USER GUIDE](#) · [ABOUT](#)

Base editor / cell type: BE4, mES

Amino acid frame: **None** ▼

Base editing outcomes among edited reads: DNA sequence

1 5 10 15 20

ATGAGGGACCTACTACGCG

CCTGCTTAAGCGTCTGTTAAATGAGGACCTACTACGAGCAGCAGGGGGC 37.8%

CCTGCTTAAGCGTCTGTTAAATGAGGAGAACTACTACGAGCAGCAGGGGGC 17.3%

CCTGCTTAAGCGTCTGTTAAATGAGGACCTACTACGAGCAGCAGGGGGC 8.9%

CCTGCTTAAGCGTCTGTTAAATGAGGACCTACTACGAGCAGCAGGGGGC 4.3%

CCTGCTTAAGCGTCTGTTAAATGAGGACAACTACTACGAGCAGCAGGGGGC 3.9%

CCTGCTTAAGCGTCTGTTAAATGAGGAGCTACTACGAGCAGCAGGGGGC 3.3%

CCTGCTTAAGCGTCTGTTAAATGAGGAGCTACTACGAGCAGCAGGGGGC 2.2%

CCTGCTTAAGCGTCTGTTAAATGAGGACCTACTACGAGCAGCAGGGGGC 1.9%

CCTGCTTAAGCGTCTGTTAAATGAGGAGCTACTACGAGCAGCAGGGGGC 1.7%

CCTGCTTAAGCGTCTGTTAAATGAGGAGCTACTACGAGCAGCAGGGGGC 1.6%

[Download table of predictions](#)

Practical

- > Download and install BEstimate
- > How run run it
- > Examine the output files

BEstimate: a computational tool for the design and interpretation of CRISPR base editing experiments

 Cansu Dinçer,  Bo Fussing,  Mathew J. Garnett,  Matthew A. Coelho

doi: <https://doi.org/10.1101/2025.05.19.654892>

<https://github.com/CansuDincer/BEstimate>